

5.8.9

EE25BTECH11011 - Navya

Question:

The area of a rectangle gets reduced by 9 square units, if its length is reduced by 5 units and breadth is increased by 3 units. If we increase the length by 3 units and the breadth by 2 units, the area increases by 67 square units. Find the dimensions of the rectangle

Solution:

Let the original length and breadth of the rectangle be l and b , respectively.

From the problem:

$$(l - 5)(b + 3) = lb - 9 \quad (1)$$

$$\Rightarrow 3l - 5b = 6 \quad (2)$$

$$(l + 3)(b + 2) = lb + 67 \quad (3)$$

$$\Rightarrow 2l + 3b = 61 \quad (4)$$

Equations (2) and (4) form the system:

$$\begin{pmatrix} 3 & -5 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} l \\ b \end{pmatrix} = \begin{pmatrix} 6 \\ 61 \end{pmatrix} \quad (5)$$

With the augmented matrix followed by row reduction:

$$\left(\begin{array}{cc|c} 3 & -5 & 6 \\ 2 & 3 & 61 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - \frac{2}{3}R_1} \left(\begin{array}{cc|c} 3 & -5 & 6 \\ 0 & \frac{19}{3} & 57 \end{array} \right) \quad (6)$$

$$\xrightarrow{R_2 \leftarrow \frac{3}{19}R_2} \left(\begin{array}{cc|c} 3 & -5 & 6 \\ 0 & 1 & \frac{171}{19} \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 + 5R_2} \left(\begin{array}{cc|c} 3 & 0 & \frac{969}{19} \\ 0 & 1 & \frac{171}{19} \end{array} \right) \quad (7)$$

$$\xrightarrow{R_1 \leftarrow \frac{1}{3}R_1} \left(\begin{array}{cc|c} 1 & 0 & \frac{323}{19} \\ 0 & 1 & \frac{171}{19} \end{array} \right) \quad (8)$$

$$\therefore \begin{pmatrix} l \\ b \end{pmatrix} = \begin{pmatrix} \frac{323}{19} \\ \frac{171}{19} \end{pmatrix} = \begin{pmatrix} 17 \\ 9 \end{pmatrix} \quad (9)$$

The dimensions of the rectangle are:

Length = 17 units, Breadth = 9 units